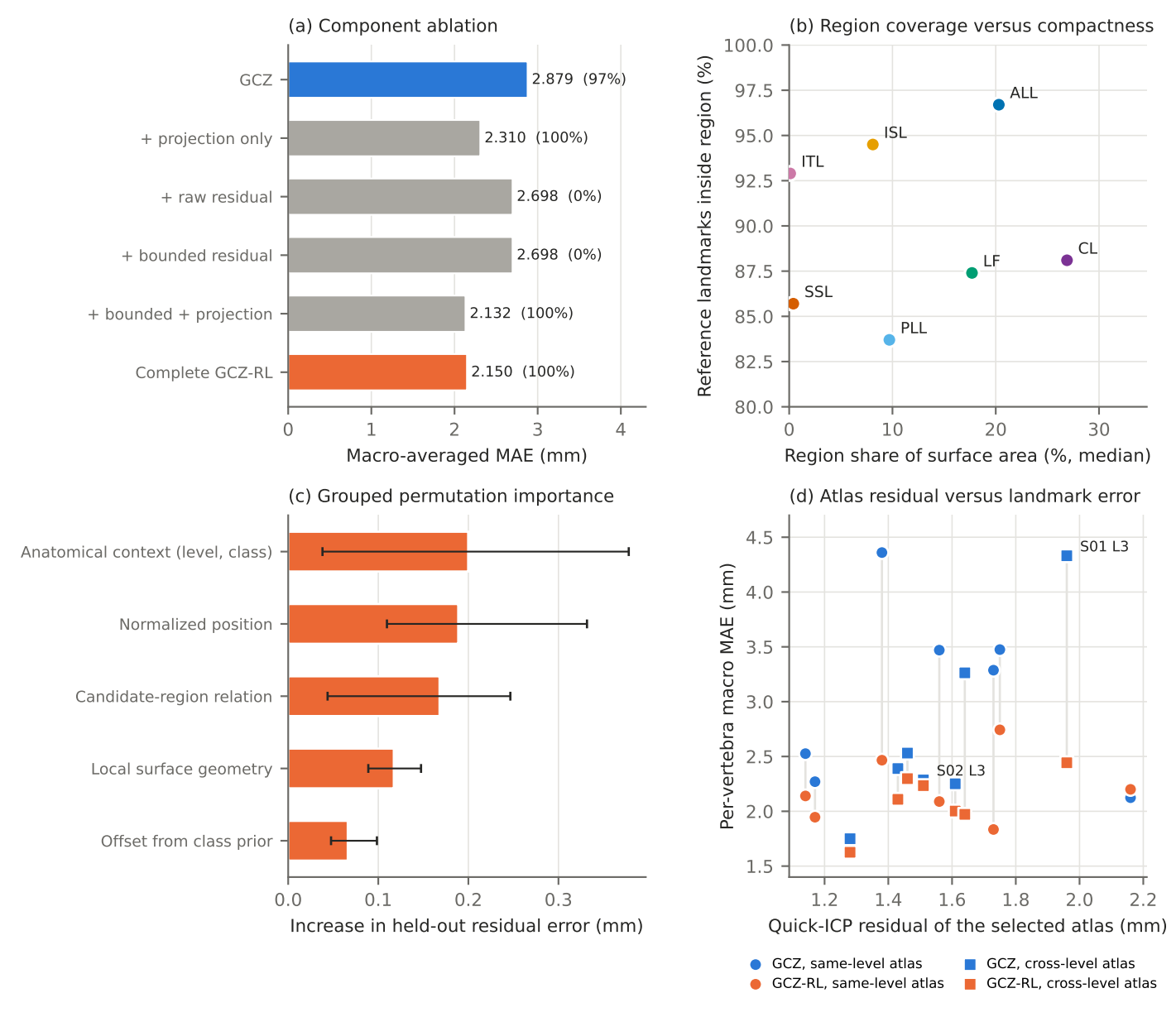
(a) Component ablation
GCZ
+ projection only
+ raw residual
+ bounded residual
+ bounded + projection
Complete GCZ-RL
2.879 (97%)
2.310 (100%)
2.698 (0%)
2.698 (0%)
2.132 (100%)
2.150 (100%)
0
1
2
3
4
Macro-averaged MAE (mm)
(b) Region coverage versus compactness
Reference landmarks inside region (%)
100.0
97.5
95.0
92.5
90.0
87.5
85.0
82.5
80.0
ALL
ISL
ITL
CL
LF
SSL
PLL
0
10
20
30
Region share of surface area (%, median)
(c) Grouped permutation importance
Anatomical context (level, class)
Normalized position
Candidate-region relation
Local surface geometry
Offset from class prior
0.0
0.1
0.2
0.3
Increase in held-out residual error (mm)
(d) Atlas residual versus landmark error
Per-vertebra macro MAE (mm)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
S01 L3
S02 L3
1.2
1.4
1.6
1.8
2.0
2.2
Quick-ICP residual of the selected atlas (mm)
GCZ, same-level atlas
GCZ, cross-level atlas
GCZ-RL, same-level atlas
GCZ-RL, cross-level atlas